

AI Security and Privacy

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Topics

Differences between AI for security and security of AI

Advantages and disadvantages of AI in cybersecurity in case studies.

Data privacy ethical and legal issues that arise from widespread availability of Generative AI

Deepfakes: How AI correctness in GANs leads to deepfakes.



AI for Security

- **AI for Security:** AI is the tool used to enhance security
- **Security of AI:** AI is the asset being protected



⊗ AI in Cybersecurity

Cybersecurity uses a combination of technologies, processes, and policies to safeguard devices and information from threats like malware, phishing, and ransomware.

Effective cybersecurity involves multiple layers of defense, including physical security, software tools, and user awareness to mitigate risks.

How AI improves security outcomes

- Uses AI techniques to detect, prevent, and respond to threats
- Analyses large volumes of data faster than humans
- Identifies anomalies and patterns in:
 - Network traffic
 - User behaviour
 - System Logs
- Applications include:
 - Intrusion and malware detection
 - Fraud detection
 - Threat intelligence and automated incident response



The **benefits** of automating AI in **cybersecurity**:



Ongoing learning



Discovering unknown
threats



Vast data volumes



Improved vulnerability
management

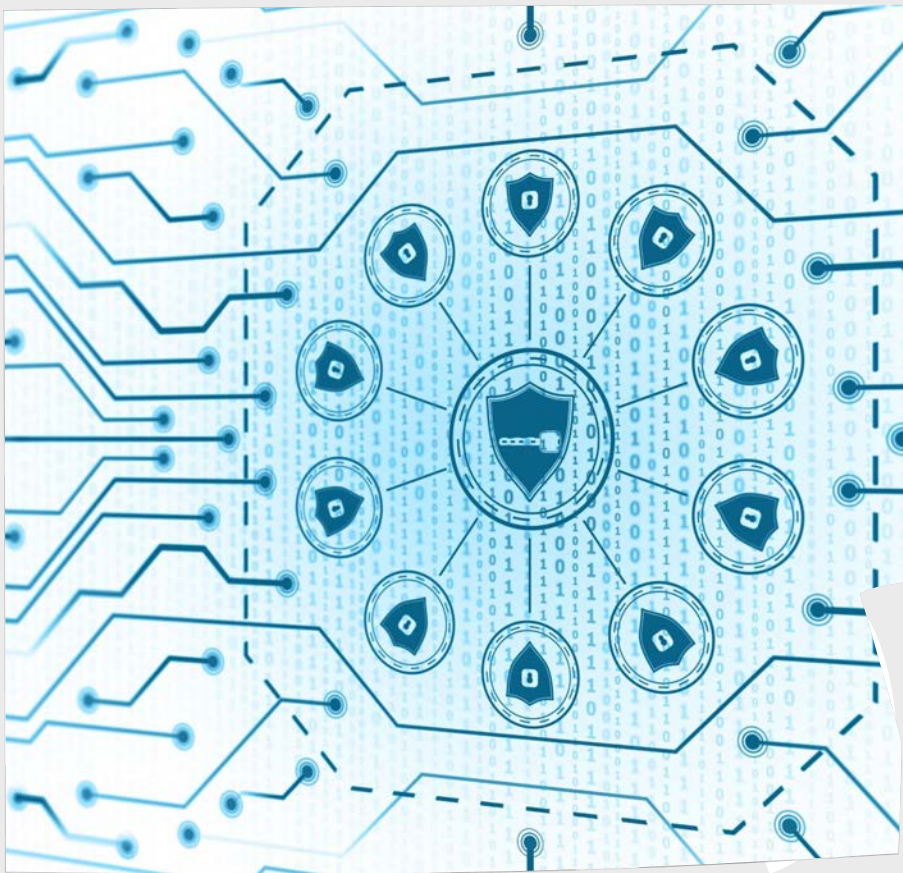


Enhanced overall
security posture



Better detection
and response

Why Cybersecurity Is Central to AI Safety



Impact of Vulnerabilities

A single cybersecurity flaw can trigger large-scale exploitation affecting millions across interconnected systems.

AI's Role in Security

AI enhances cybersecurity by automating code analysis and reducing asymmetry between defenders and attackers.

Dual-Use Risk

Vulnerability discovery is valuable for defense but risky if weaponized, illustrating dual-use challenges in AI safety.

Governance and Structural Risk

Managing AI capabilities requires governance on access and disclosure to address systemic cybersecurity risks effectively.

Traditional Cybersecurity Before Advanced AI

Reliance on Human Expertise

Traditional cybersecurity depended heavily on skilled engineers manually auditing complex source code.

Limitations of Automated Tools

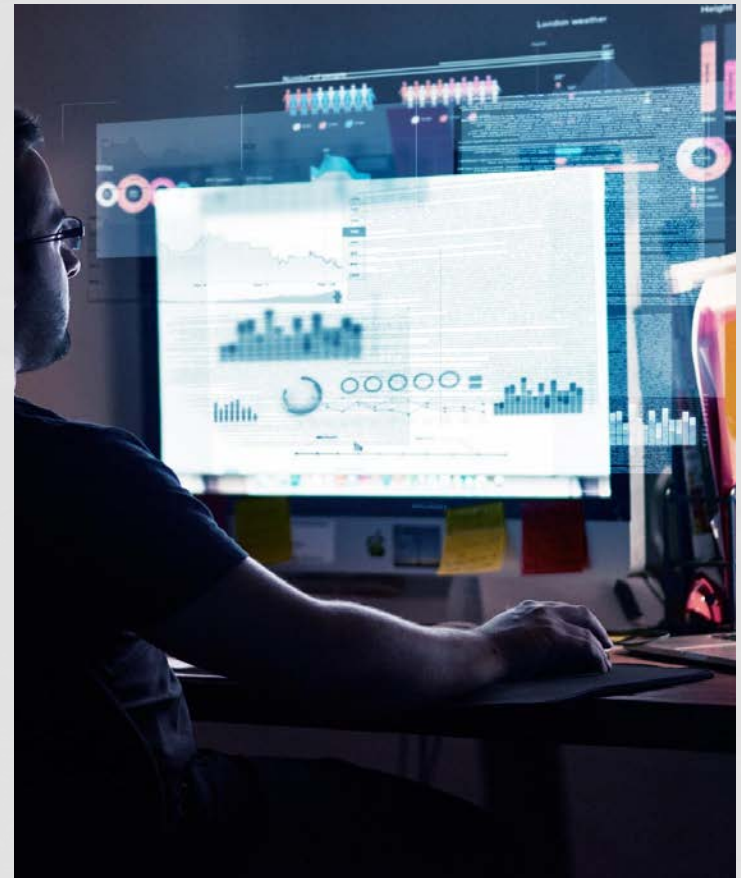
Static analysis and fuzz testing improved detection but missed subtle and complex vulnerabilities.

Challenges and Vulnerability Backlogs

Traditional methods left many vulnerabilities undiscovered for years, increasing security risks.

Need for AI Integration

AI emerged to address scalability and complexity limits inherent in traditional security approaches.



What AI Adds to Cybersecurity Capabilities

Advanced AI Reasoning

Modern AI analyzes large codebases like human experts, enabling system-level reasoning beyond pattern matching.

Vulnerability Discovery

AI uncovers hidden and complex vulnerabilities missed by traditional automated tools, improving threat detection.

Double-Edged Impact

AI can generate exploits useful for defense but also lowers barriers for attackers, posing safety challenges.

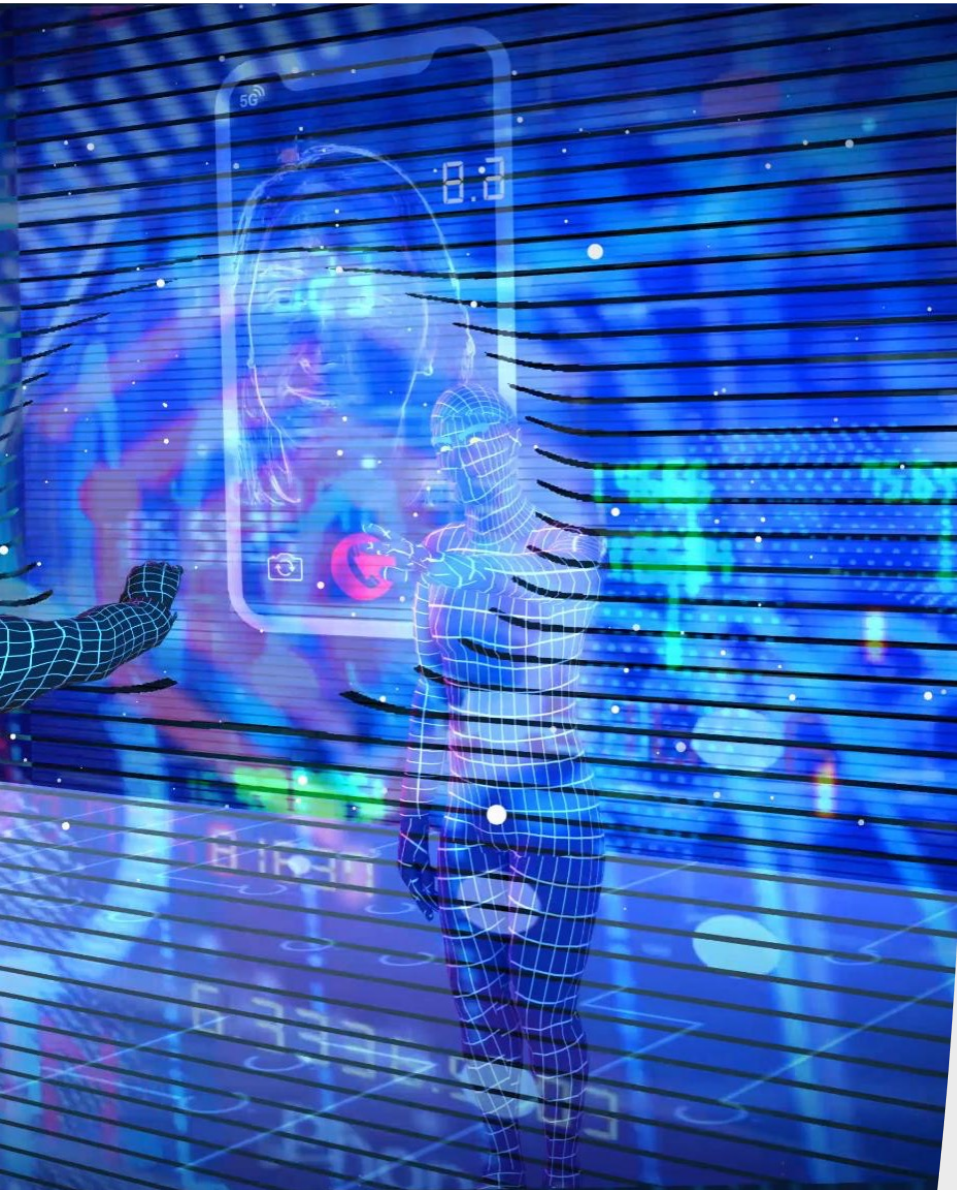
Accelerated Cybersecurity Tempo

AI compresses the time between vulnerability discovery and exploitation, necessitating integrated safety measures.





AI for Security Case Studies



Case Study: Google Project Zero and DeepMind's Big Sleep

AI-Augmented Vulnerability Discovery

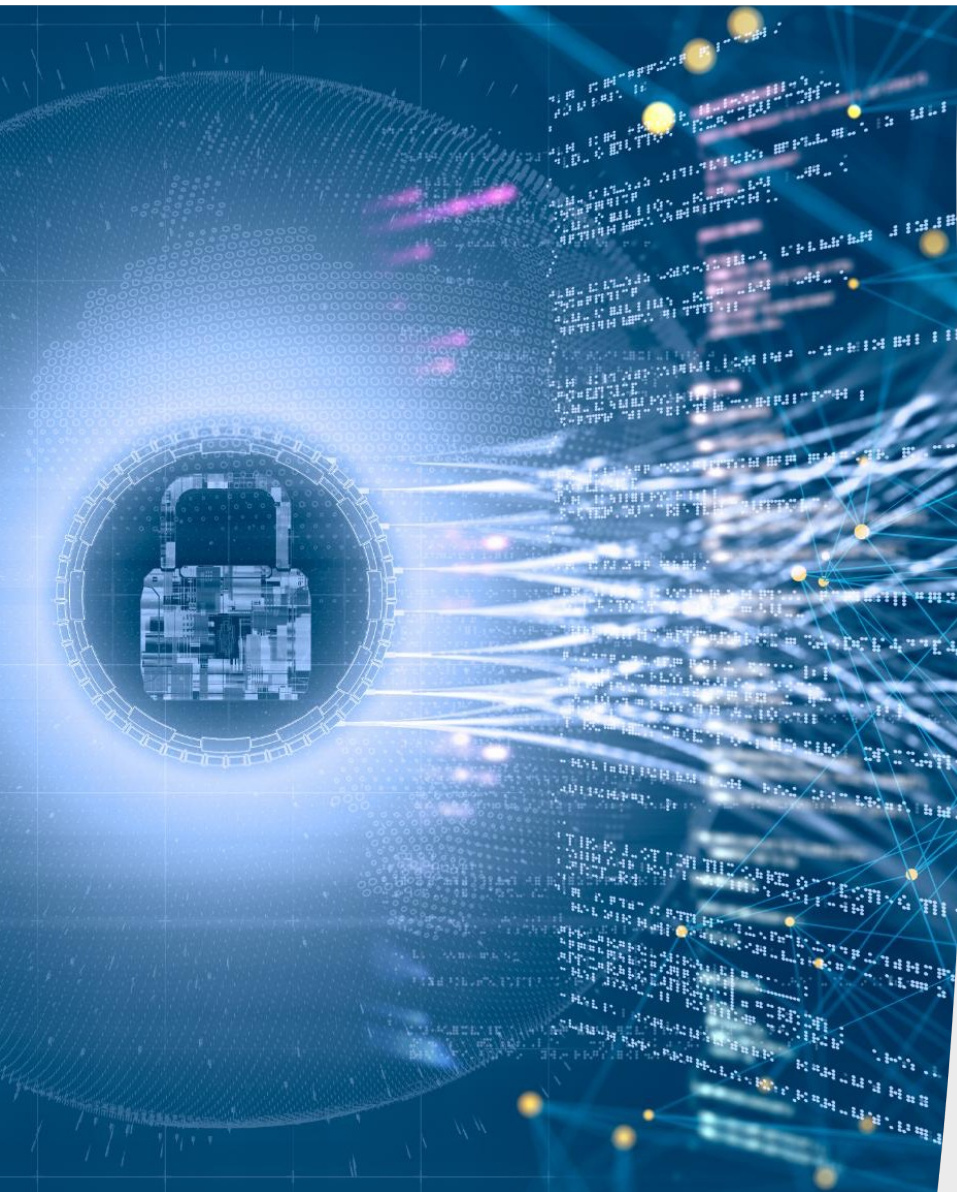
Big Sleep enhances human researchers by mimicking hypothesis-driven code analysis and testing in secure environments.

Collaborative Interdisciplinary Approach

The project combines expert vulnerability research with advanced AI, showcasing the value of interdisciplinary collaboration.

Safety and Responsible Use

Big Sleep operates in controlled research contexts emphasizing defensive use and responsible vulnerability disclosure.



Big Sleep in Action: Discovering a Real Zero-Day

Autonomous Zero-Day Discovery

Big Sleep autonomously found a subtle memory safety flaw in SQLite, previously undetected by traditional testing.

Significance for AI Safety

This discovery shows AI can exceed traditional security tools but requires careful ethical governance.

Coordinated Response and Governance

Success depended on governance ensuring the flaw was responsibly disclosed and swiftly patched.

Why Big Sleep Is a Positive AI Safety Example



Proactive Vulnerability Detection

Big Sleep was used proactively to identify vulnerabilities before attackers could exploit them.

Restricted Trusted Access

Access to Big Sleep was limited to trusted security teams to minimize misuse risks.

Human Accountability

Humans were responsible for validation, disclosure, and remediation within a safety framework.

Systemic Benefit

Patching widely used components improved security for many downstream users.

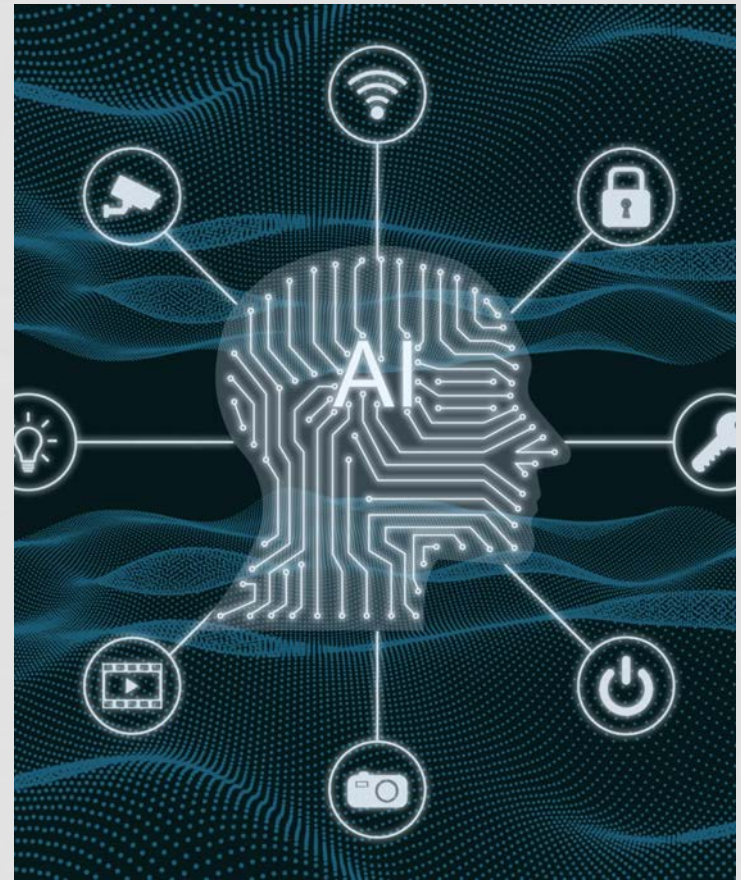
Case Study: Anthropic's Project Glasswing

Claude Mythos demonstrated **advanced ability** to discover and chain software vulnerabilities effectively.

Project Glasswing **limits model access** to trusted organizations to defensively identify and fix vulnerabilities.

Anthropic shifted focus from alignment alone to **containment** and controlled deployment for safer AI use.

Project Glasswing emphasizes **defender-first approach**, differing in scale and coordination from similar projects.



Article in The Conversation about Mythos

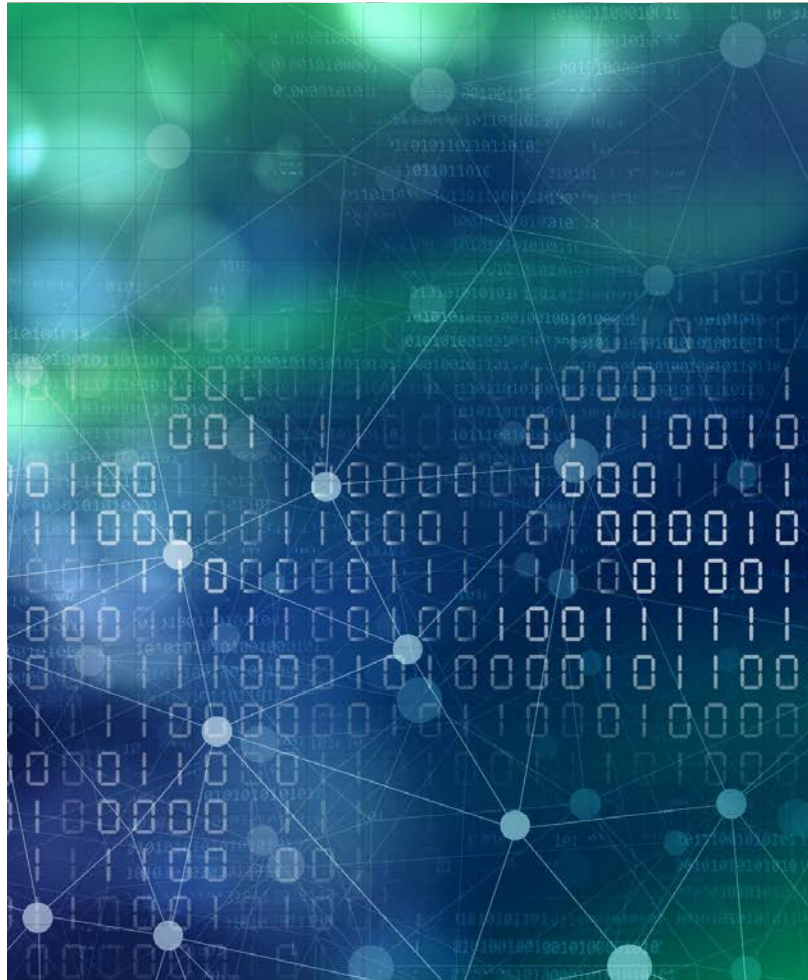
13 April 2026

<https://theconversation.com/claude-mythos-and-project-glasswing-why-an-ai-superhacker-has-the-tech-world-on-alert-280374>



Claude Mythos and Project Glasswing: Why an AI Superhacker Has the Tech World on Alert

EXPLORING
CYBERSECURITY THREATS
AND AI INNOVATIONS'
GROWING IMPACT



What Is Project Glasswing?

Defensive Cybersecurity Initiative

Project Glasswing uses AI to proactively identify and fix cybersecurity vulnerabilities before attackers exploit them.

Collaborative Vulnerability Sharing

Findings from software scans are responsibly shared with maintainers to develop timely security patches.

Collective Digital Defense

Glasswing fosters cooperation among organizations to strengthen the overall digital ecosystem against AI threats.

Who Is Involved in Glasswing



Tech Industry Leaders

Major tech companies form the backbone of Glasswing, addressing systemic digital security risks worldwide.

Open-Source Organizations

Open-source groups like the Linux Foundation help protect essential software maintained by small teams.

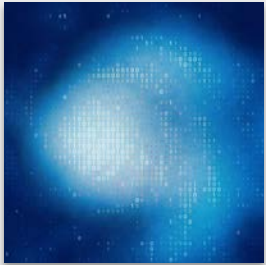
Financial Institutions

Financial firms such as JPMorgan Chase participate to secure critical financial digital systems.

Cross-Sector Collaboration

Glasswing highlights the importance of collaboration across sectors to combat advanced AI-driven cyber threats.

The AI Superhacker Narrative



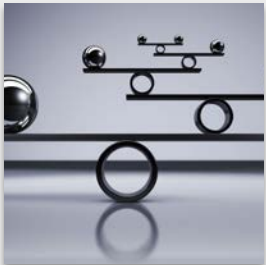
Unreleased AI Superhacker

Claude Mythos is a powerful unreleased AI that uncovers deep software vulnerabilities across major systems and browsers.



Restricted Access and Defense

Anthropic restricts Mythos's access and directs it toward defensive cybersecurity with Project Glasswing to reduce risks.



Balancing Cybersecurity Risks

Mythos shifts the balance between attackers and defenders, highlighting trust, governance, and digital resilience challenges.

What Is Claude Mythos?

Advanced Code Analysis

Claude Mythos excels in analyzing code for bugs and transforming them into functional exploits, enabling deeper cybersecurity insights.

Wide Vulnerability Discovery

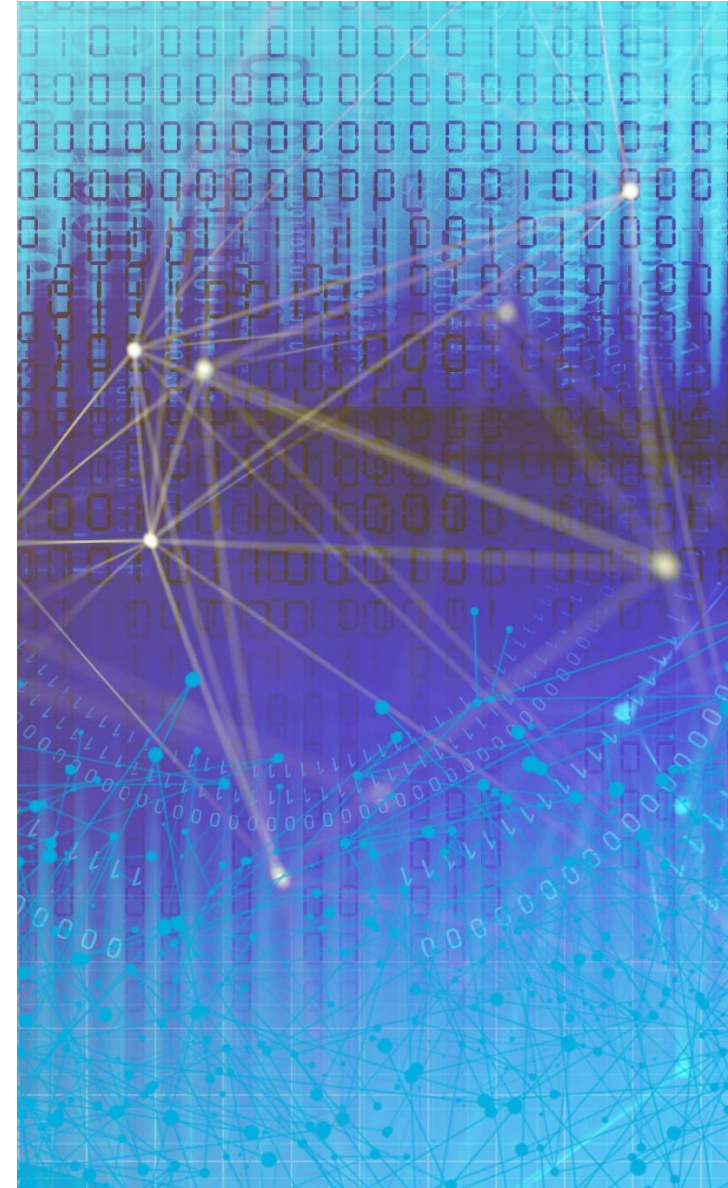
Mythos has identified vulnerabilities across major operating systems and web browsers, surpassing previous AI capabilities.

Restricted Access Model

The model is a gated AI tool available only to trusted partners, reflecting concerns about its powerful offensive potential.

Implications for AI Governance

Mythos challenges current AI safety and cybersecurity paradigms, emphasizing the need for thoughtful governance and control.



Why the Tech World Is Alarmed

AI Accelerates Cyberattacks

AI like Mythos can be exploited to quickly uncover hidden vulnerabilities that humans and traditional tools miss.

Scale and Efficiency of AI

Unlike human hackers, AI analyzes vast codebases rapidly, lowering effort required for sophisticated attacks.

Widespread Impact Risk

Vulnerabilities in common software components risk cascading effects across millions of systems worldwide.

Urgency for Countermeasures

Cybersecurity must evolve to address AI-driven threats before offensive capabilities surpass defenses.





Examples of Major Vulnerabilities Found

OpenBSD* Long-Undetected Flaw

Mythos discovered a 27-year-old vulnerability in OpenBSD, despite its strong security reputation.

Ffmpeg Hidden Weakness**

A 16-year-old flaw in FFmpeg was found by Mythos, highlighting hidden risks in widely used software.

Linux* Kernel Exploit Chain**

Mythos chained multiple Linux kernel vulnerabilities to gain full system control, demonstrating advanced attack methods.

* OpenBSD is a free, secure, and portable Unix-like operating system based on 4.4BSD, focusing on proactive security, code correctness, and integrated cryptography.

** FFmpeg is a free, open-source software suite and industry-standard command-line tool used to record, convert, stream, and edit multimedia files.

*** Linux is a free, open-source operating system kernel created by [Linus Torvalds](#) in 1991, which acts as the core of a computer system, managing hardware resources like CPU, memory, and devices

Can Mythos Act Autonomously?

Low Risk of Autonomous Harm

Mythos is unlikely to act independently in harmful ways or develop its own malicious goals.

Risk of Human-Directed Misuse

The main danger lies in how effectively Mythos can perform harmful tasks when directed by humans.

Importance of AI Governance

Access control, auditing, and usage restrictions are crucial to prevent unethical AI deployment.

Controlled Deployment Environment

Project Glasswing provides a controlled setting to ensure AI is used for ethical, defensive purposes.

Why Anthropic Is Keeping Mythos Restricted

Risk of Premature Release

Anthropic views Mythos's advanced capabilities as too risky for public release without strong safeguards.

Preventing Cybersecurity Threats

Restricting Mythos helps stop malicious actors from exploiting vulnerabilities and shifting cybersecurity balance.

Responsible AI Stewardship

Anthropic treats Mythos as high-risk technology requiring careful control rather than a commercial product.

Challenging Industry Norms

The decision reflects a philosophy favoring restraint and coordination over rapid, open deployment.



Why This Development Matters



AI Advances in Cybersecurity

AI like Mythos enables systematic vulnerability discovery, challenging traditional cybersecurity defenses.

Reassessing Security Strategies

Automated vulnerability discovery demands new defense strategies and raises concerns about access and coordination.

Dual-Use AI Challenges

AI's power poses both risks and defense benefits, requiring balanced governance and ethical management.

From Tools to Capabilities: A Governance Shift

Governance Beyond Tools

AI governance must focus on underlying capabilities, not just tools and outputs, to manage risks effectively.

Limitations of Post-Hoc Mitigation

Traditional software fixes harms after release, but AI vulnerabilities require proactive governance measures.

Central Governance Mechanisms

Access control, capability evaluation, and pre-release decision-making are essential for safe AI deployment.

Cybersecurity as a Model

Cybersecurity highlights the need for strict governance as misuse consequences can be immediate and severe.



Implications

Rapid AI Advances in Cybersecurity

AI capabilities in cybersecurity are progressing faster than expected, surpassing many expert predictions.

Responsible AI Stewardship

Limiting AI access demonstrates responsible development contrasting with typical fast deployments in the industry.

Proactive Cyber Defense Models

Initiatives like Project Glasswing try to align AI tools with defensive cybersecurity goals to manage emerging risks.

Governance Shapes Technological Impact

The impact of AI hinges on governance decisions around access, collaboration, and responsibility in cybersecurity.



AI, Cybersecurity, Risk, and Safety

AI's Role in Cybersecurity

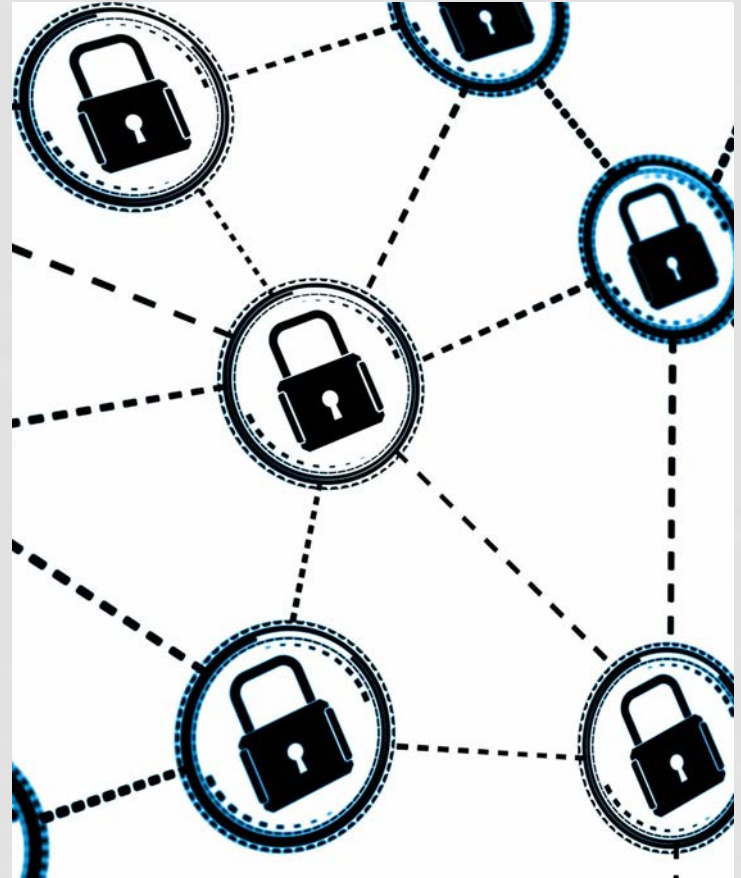
AI has become central in cybersecurity, reshaping risk and enhancing defense capabilities at a global scale.

Dual-Use Risk of AI

The same AI systems can improve security or accelerate cyber attacks if misused or poorly governed.

Cybersecurity in the AI Safety Context

Cybersecurity grounds AI safety concepts, linking technical excellence with ethical responsibility and risk trade-offs.



Design Principles for Safe Defensive AI



Restricted Access and Control

Safe AI design mandates restricted access to powerful systems to prevent misuse and unauthorized exploitation.

Human Oversight and Accountability

Human supervision at critical AI decision points ensures accountability and corrects potential errors or risks early.

Responsible Disclosure Norms

Adhering to disclosure norms promotes transparency and responsible communication of AI vulnerabilities and updates.

Avoiding Unsafe Anti-Patterns

Unsafe practices like unrestricted public release or lack of rollback risk undermine AI security and trust.

Implications for Engineers and Researchers

Beyond Technical Skills

Engineers must combine technical expertise with judgment to decide what AI developments should be built or restricted.

Ethical and Safety Dilemmas

Discovering vulnerabilities or advanced methods poses ethical challenges that require careful consideration and responsibility.

Responsible Practice

Practitioners should consider downstream impacts, access control, and collaborate with institutions to ensure safe AI use.

Professionalism Through Restraint

Recognizing when to refuse or restrain AI deployment is a mark of professionalism and ethical responsibility.



Implications for AI Safety Education

Comprehensive AI Safety Training

Students must learn to build models and understand failure modes, risk analysis, and governance for safe AI deployment.

Cybersecurity as Teaching Domain

Cybersecurity offers a concrete domain where AI safety lessons on misuse consequences are accessible and accepted.

Integration of Perspectives

AI safety education should blend technical, institutional, and ethical views to prepare students for restricted system environments.

Use of Real Case Studies

Employing real-world cases enables practical frameworks for responsible AI decision-making beyond abstract theory.





AI Privacy

Understanding AI Privacy

Purpose:

Protecting individuals' data used in AI systems from misuse or unintended disclosure

Concerns:

- Data leakage during training/testing
- Identity privacy of individuals in data sets
- Risk of re-identification from generated outputs



Challenges to Privacy in AI

Data Collection and Use

- Large datasets may contain personal information
- Data sourced from public or private sources without consent

Model Training Risks

- Embedding sensitive data in model weights
- Extracting personal data from trained models (model inversion attacks)
- Generation of Harmful Content



Ethical and Legal Considerations

Ethical Challenges:

- Consent for using individuals' images/data
- Accountability for malicious use

Regulations & Laws:

- General Data Protection Regulation (GDPR)
 - Enacted in the European Union (EU) on May 25, 2018.
 - One of the strictest and most influential privacy laws worldwide.
 - Applies to organizations anywhere, if they target or collect data related to EU residents.
- California Privacy Rights Act (CPRA) governing data privacy rights passed in 2018
- The Lei Geral de Proteção de Dados, or LGPD, is a data privacy regulation passed by the National Congress of Brazil in 2018.



GDPR Core Principles

- **Lawfulness, Fairness, and Transparency:** Data processing must be legal, fair, and clear to the individual.
- **Purpose Limitation:** Data collected only for specified, explicit, and legitimate purposes.
- **Data Minimization:** Only necessary data should be collected.
- **Accuracy:** Personal data must be kept accurate and up-to-date.
- **Storage Limitation:** Data should only be retained as long as necessary.
- **Integrity and Confidentiality:** Data must be processed securely to prevent unauthorized access/loss.



GDPR: Key Rights for Data Subjects

- **Right to be Informed:** Know how your data is used.
- **Right of Access:** Request a copy of your data.
- **Right to Rectification:** Correct inaccurate data.
- **Right to Erasure** ("Right to be Forgotten"): Request deletion of your data under specific conditions.
- **Right to Data Portability:** Transfer your data to another provider.
- **Right to Object:** Stop your data from being used for specific purposes like marketing.





GDPR vs CCPA vs LGPD: Key Rights Comparison

GDPR Comprehensive Rights

GDPR provides extensive rights including data access, correction, erasure, restriction, objection, portability, and automated decision protections.

CCPA Consumer Focus

CCPA emphasizes consumer transparency, choice, data deletion, opt-out of data sale, and non-discrimination protections.

LGPD Alignment and Transparency

LGPD aligns with GDPR, focusing on data access, correction, anonymization, portability, consent revocation, and clear transparency.

Data Privacy Rights Comparison

GDPR Data Subject Rights	CCPA Data Subject Rights	LGPD Data Subject Rights
Right of access	Right to know	Right of access
Right of rectification	Right to delete	Right to confirmation of the existence or the processing
Right to erasure ("right to be forgotten")	Right to opt-out of sale or sharing personal information	Right to correct
Right of data portability	Right to correct	Right to anonymize, block, or delete
Right to object	Right to limit the use and disclosure of sensitive personal information	Right to delete personal data that the individual has previously given consent to processing
Right to restriction of data processing	Right to be notified	Right to data portability
Rights related to automated decision-making such as profiling	Right of non-discrimination	Right to being informed about 3rd parties the data controller is sharing or selling data to
Right to withdraw consent		Right to be informed about the possibilities and consequences of denying consent
		Right to revoke consent



Deepfakes as a Privacy Concern

What are Deepfakes?

- AI-generated realistic images or videos of people
- Created using generative models, often GANs

Privacy Concerns:

- Unauthorized use of individuals' likenesses
- Potential defamation, blackmail, misinformation
- Difficulty in removing deepfake data from the internet

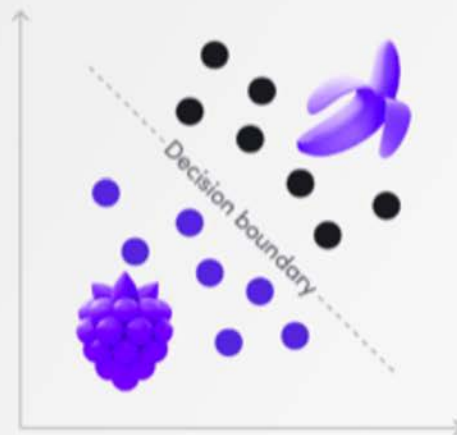




Adversarial Networks

Discriminative

Classify or label data point
as banana or berry



Generative

Produce a new data point that looks
like bananas or berries



www.miquido.com



The Two Competing Models

GANs consist of two neural networks trained together:

Generator

- Produces synthetic data (images, audio, video, text)
- Goal: create outputs that look *real*

Discriminator

- Evaluates inputs and decides whether they are *real* (from training data) or *fake* (from the generator)
- Goal: correctly identify fakes



The Adversarial Training Loop

- The **generator makes errors** (produces outputs the discriminator can spot)
- The **discriminator detects those errors**
- Feedback is used to **reduce errors in the generator**
- This cycle repeats thousands or millions of times

Over time:

- The discriminator becomes *less able to tell the difference*
- The generator's outputs become *increasingly realistic*

When the discriminator can no longer reliably distinguish real from fake, the generator is considered **successful**.



Deep Fakes as a By-Product of Error Reduction

Deep fakes emerge because GANs:

- Learn fine-grained patterns in faces, voices, gestures, and speech
- Reduce visual or auditory artifacts that reveal manipulation
- Optimise specifically for **human perceptual realism**

As error approaches zero:

- Fake faces look real
- Synthetic voices sound authentic
- Fabricated videos appear convincing



Risks: Misinformation (Unintentional Harm)

GAN-generated content can be:

- Used mistakenly as real
- Shared without malicious intent
- Believed because it *looks and sounds right*

Examples:

- Fake images used in news reporting
- Altered videos shared on social media
- Synthetic audio misattributed to real people



Risks: Disinformation (Deliberate Harm)

Because adversarial networks can produce **high-credibility fakes**, they are powerful tools for intentional deception:

- Political manipulation
- Fake evidence or fabricated statements
- Impersonation and scams
- Erosion of trust in genuine media
- Here, low error equals **high persuasive power**.

The Deeper Risk: Loss of Trust

As adversarial realism improves:

- People can no longer easily trust what they see or hear
- Genuine evidence can be dismissed as “just another deep fake”



Scenario: From Adversarial Correctness to Loss of Trust

A media company uses a **GAN-based video system** to generate realistic video content for fast news explainers and reconstructions.

The system was trained on thousands of hours of real news footage.

Its goal is to **reduce visual and audio errors** so generated videos are indistinguishable from real recordings.

“Correctness” is defined internally as:

The discriminator can no longer tell real footage from generated footage

During a breaking news event, the system generates a video of a public official making a short statement that *reconstructs* what was “likely said,” based on:

- Past speeches
- Voice patterns
- Facial movements

The generated video:

- Looks real
- Sounds real
- Matches known communication style



Disinformation becomes possible

Later, malicious actors realise:

- They can generate similar videos with targeted messages
- The system's strength—minimal error—makes deception easy

They deliberately:

- Release fabricated statements
- Time releases to influence public opinion
- Rely on realism to bypass scepticism

This becomes **disinformation**:

- False content
- Shared intentionally
- Designed to mislead



Trust Begins to Collapse

Once the deep fake is exposed:

- The company issues a correction
- But damage is already done

Public reaction shifts:

- “You can’t trust video anymore”
- Real footage is questioned
- Genuine evidence is dismissed as “probably fake”

This leads to:

- Erosion of trust in media
- Confusion about what is real
- Increased cynicism toward legitimate information



Adversarial AI “Correctness” Optimises the Wrong Thing

- Correctness = low detectable error
- Success = indistinguishability from reality
- There is no concept of truth, intent, or context

The system performs exactly as designed.



Summary

AI Enhances Cybersecurity

AI improves cybersecurity by enabling proactive vulnerability discovery and reducing systemic risk effectively.

Governance and Oversight

Effective governance including access control, oversight, and coordination is critical for maximizing AI benefits.

Beyond Alignment

Alignment alone is not enough; capability containment and institutional responsibility are essential.

